

Department Overview

- **Department Name:** Department of Textile Engineering & Design
- **Faculty:** Faculty of Engineering & Architecture
- **Accreditation:** Fully accredited by the **Pakistan Engineering Council (PEC)**.
- **Core Focus:** Encompasses the full manufacturing lifecycle of textiles—transforming raw fibers into yarn, spinning yarn into fabric, and executing garment/apparel and home textile designs. It includes processing of wool, cotton, and manmade fibers, technical textiles, color science, and environmental sustainability in manufacturing.
- **History & Impact:** Launched in 2004 as a pioneer engineering program at the Takatu Campus with an initial intake of 20 students. Strategically positioned because Balochistan is rich in raw wool production. The department introduced its postgraduate (MS) program in 2015.
- **Career Pathways:** Graduates work as technical experts, supervisors, and managers across textile processing plants, quality control units, research labs, product development divisions, and corporate management. The field also branches into specialized biomedical applications (e.g., medical filters and artificial arteries).

Vision and Mission

- **Vision:** To be recognized internationally for its quality research, educational activities, and its relationship with the environment. Its graduates will be acknowledged for the strength of their technical and practical knowledge, ability to work as a team, and their vocation of service to society.
- **Mission:** To provide professional training, promoting social responsibility, values, and a progressive research culture among the students of Textile Engineering, making them capable of providing solutions to complex engineering problems, preserving the environment, and contributing to the sustainable development of Pakistan and the society.

1. BS Textile Engineering (Engineering Track)

Admission Requirements

- **Academic Background:** F.Sc (Pre-Engineering) / ICS with Mathematics & Physics from a recognized board (or equivalent qualification) with at least **60% marks**, OR a Diploma of Associate Engineer (DAE) in a relevant discipline with a minimum of **60% marks**.
- **Testing:** Success in the NTS Test is required.

Degree Requirements

- **Total Credit Hours:** 136

- **Total Courses:** 50
- **Minimum Academic Standing:** CGPA $\geq$ 2.0

Program Educational Objectives (PEOs)

1. Implement real-world applications making use of fundamental engineering knowledge, designs, and managerial skills gained during the textile engineering program.
2. Generate innovative and sustainable solutions to complex engineering problems contributing to the textile industry.
3. Promote themselves through communication and interpersonal skills, career management strategies, and the ability to deal with technological, socio-economical, environmental, and ethical challenges.

Curriculum Plan (Program Schema by Semester)

Note: Credit hour listings follow the format (Theory - Lab).

Semester	Course Code	Course Title	Prerequisite	Credit Hours
1st	MATHP-212 / CHE-102 & L	Linear Algebra & Geometry / Applied Chemistry	None / None	3-0 / 3-1
	PHY-205 & L / TXTE-202	Applied Physics / Textile Raw Materials-I	None / None	3-1 / 3-0
	HUM-163 / TXTE-103	Functional English (English-I) / Intro to Textile Eng.	None / None	2-0 / 2-0
2nd	MATHP-167 / ENGG-201L	Calculus-I / Engineering Drawing & Graphics	None / None	2-0 / 0-1

	ENGG-218 & L / TXTE-313	Engineering Mechanics / Textile Raw Materials-II	None / None	2-1 / 3-0
	HUM-101 / CS- 105TXTE & L	Islamic Studies / Introduction to Computing	None / None	2-0 / 1-1
	HUM-268 / MATHA-546 & L	Communicatio n Skills (Eng-II) / Thermodynami cs	None / None	2-0 / 2-1
3rd	MATHP-267 / TXTE-401 & L	Calculus-II / Fibre Science	None / None	2-0 / 3-1
	ENGG-219 / HUM-102	Machine Design / Pakistan Studies	None / None	2-0 / 2-0
	TXTE-302 & L / TXTE-231 & L	Spinning Preparatory Process / Fabric Manufacturing -I	None / None	2-1 / 2-1
4th	STAT-103 / EE- 209 & L	Statistics and Probability / Electrical & Electronic Eng.	None / None	3-0 / 2-1

	HUM-207	Professional Ethics	None	2-0
	TXTE-321 & L / TXTE-331 & L	Yarn Production Process / Fabric Manufacturing -II	None / None	3-1 / 3-1
5th	MGMT-304 / HUM-265	Organizational Behavior / Tech Writing & Pres. Skills	None / Eng-I	2-0 / 3-0
	TXTE-422 & L / TXTE-431 & L	Spinning Planning & Production / Weave Designs	None / None	2-1 / 3-1
	TXTE-304 & L / TXTE-124 & L	Pre-treatment in Textile Wet Processing / Garment Fund.	None / None	2-1 / 2-1
6th	MGMT-314 / EE-343 & L	Operations & Project Management / Process Systems	None / None	2-0 / 2-1
	TXTE-456 & L	Textile Dyes & Dyeing	Applied Chemistry	2-1

	TXTE-455 & L / TXTE-453	Anthropometry & Clothing Construction / Tech Textiles	None / None	2-1 / 2-0
	TXTE-311 / TXTE-402	Textile Utilities & Services / Color Science	None / None	2-0 / 2-0
	—	Mandatory Industry Internship (4- 6 weeks)	—	Non-Credit
7th	ECON-105 / TXTE-452	Engineering Economics / Merchandizing	None / None	2-0 / 2-0
	TXTE-441L / CS-203L	Design Project-I / Computer Application in Textiles	None / Computing	0-3 / 0-2
	TXTE-351 & L / TXTE-352 & L	Mechanism of Garment Machines / Sewn Production Eng.	None / None	2-1 / 2-1
	TXTE-432	Denim Technology	None	2-0
8th	MGMT-201 /	Entrepreneurs	None / None	2-0 / 3-1

	TXTE-444 & L	hip / Textile Testing & Quality Control		
	ENGG-323 & L	Environmental Issues & Control	None	2-1
	TXTE-457 & L / TXTE-442L	Textile Printing & Finishing / Design Project-II	None / None	3-1 / 0-3

2. BS Fashion & Textile Design (Non-Engineering Track)

Admission Requirements

- **Academic Background:** Intermediate certification (F.Sc. / ICS / F.A.) from any recognized board or an equivalent qualification with at least **45% marks**.
- **Testing:** Must pass both the central NTS Test and a localized **Free Hand Drawing Test**.

Degree Requirements

- **Total Credit Hours:** 136
- **Total Courses:** 64
- **Minimum Academic Standing:** CGPA $\geq$ 2.0